

JABstem

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Past Papers

Nat 5

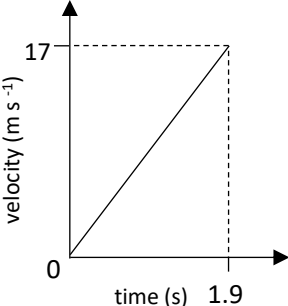
Physics

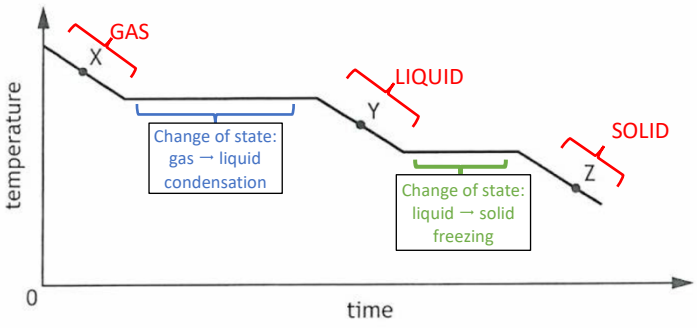
2026 Marking Scheme

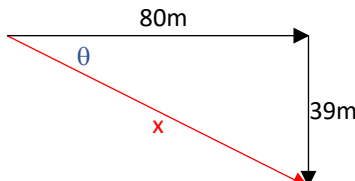
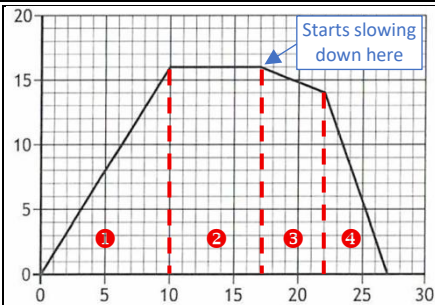
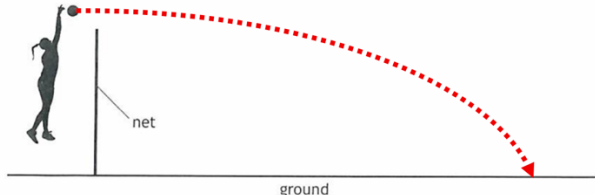
Grade Awarded	Mark Required		% candidates achieving grade
	/125	%	
A	83+	66.4%+	%
B	68+	54.4%+	%
C	54+	43.2%+	%
D	39+	31.2%+	%
No award	<39	<31.2%	%

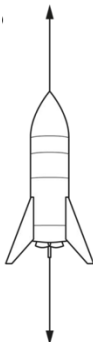
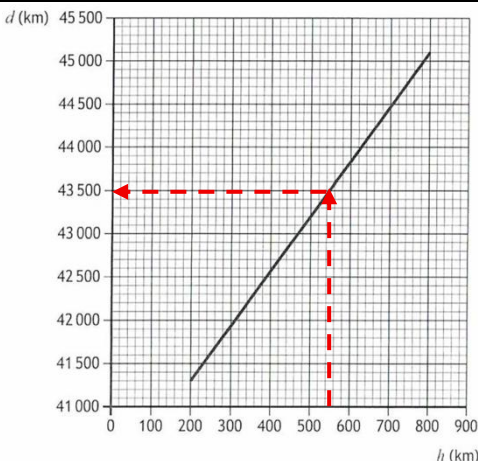
Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	14.2 /25	37.8 /75	17.3 /25

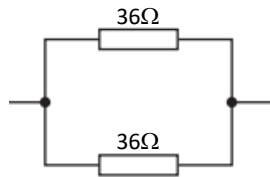
2026 Nat5 Physics Marking Scheme

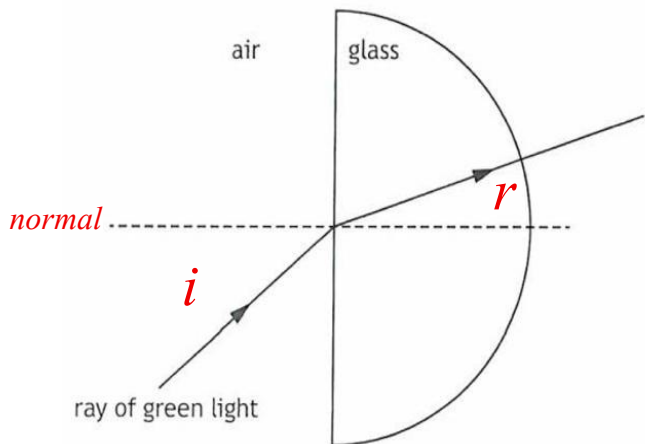
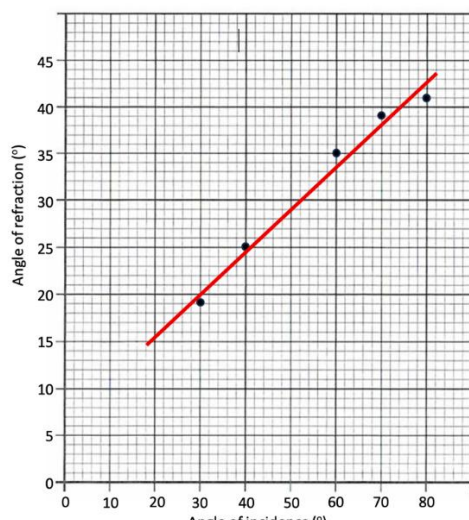
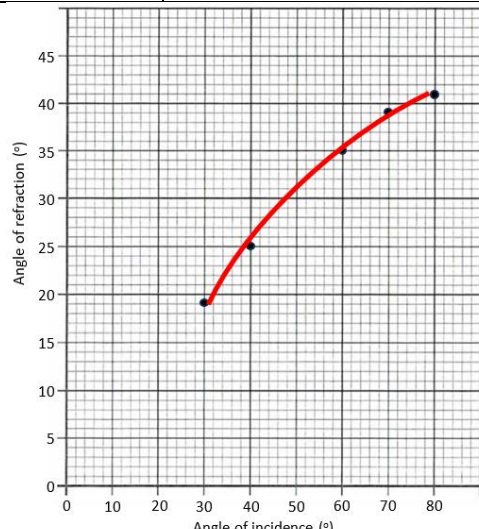
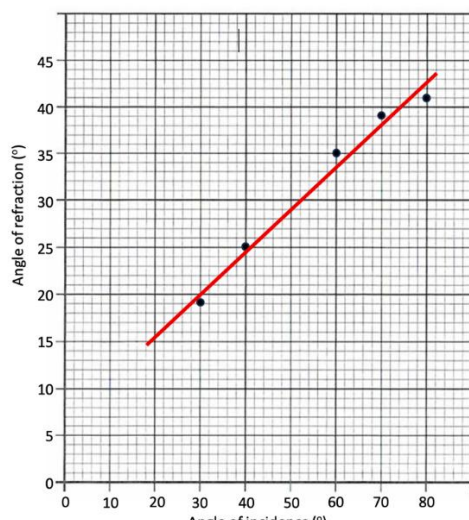
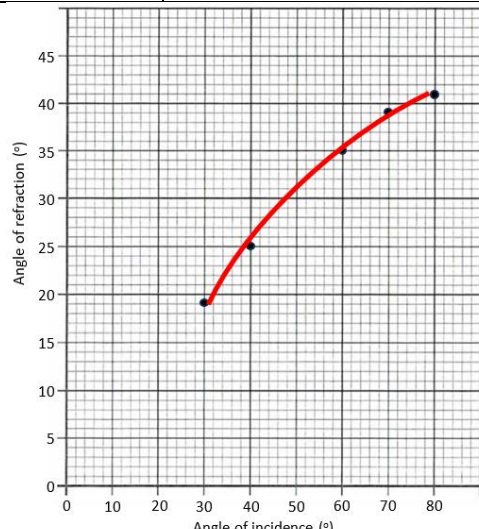
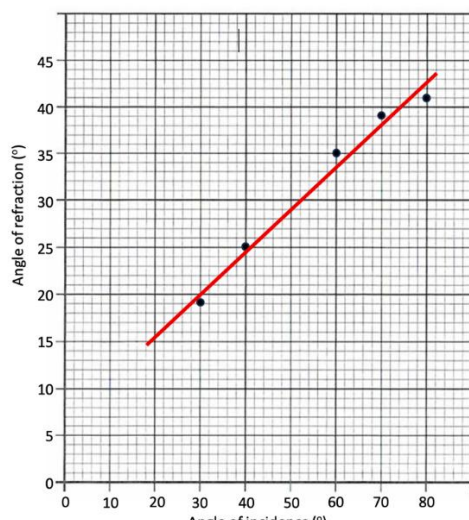
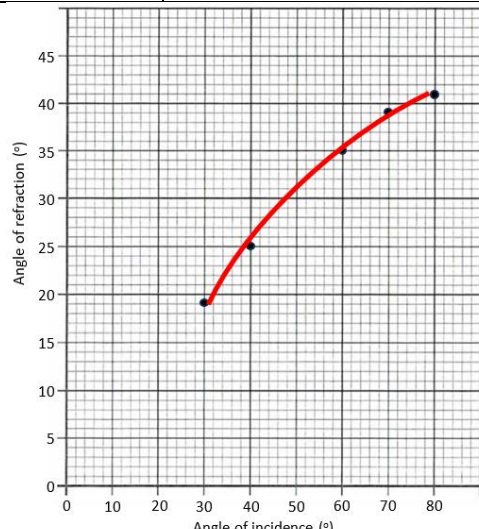
Question	Answer	% Correct	Physics Covered					
1	A	81	Scalar Quantity	energy	speed	distance	time	mass
			Vector Quantity	force	velocity	displacement	acceleration	weight
2	D	69	$d = 0.052 \text{ m s}^{-1}$ $t = 0.037 \text{ s}$ $v = \frac{d}{t} = \frac{0.052}{0.037} = 1.42 \text{ m s}^{-1}$ $v = ?$					
3	D	24			distance travelled = area under graph = $\frac{1}{2} \times 17 \times 1.9$ = 16.15m			
4	C	69	$E_k = 0.90 \text{ J}$ $m = 0.20 \text{ kg}$ $v = ?$ $E_k = \frac{1}{2} m v^2$ $0.90 = \frac{1}{2} \times 0.20 \times v^2$ $v^2 = 9$ $v = \sqrt{9}$ $v = 3.0 \text{ m s}^{-1}$					
5	B	93	Word	X	Y	Z		
			Statement	The Sun is an example of a	A natural satellite of a planet is called a	A group of planets that orbit a star is called a		
			Answer	star	moon	solar system		
6	D	24	Statement I - Correct		Statement II - Correct		Statement III - Incorrect	
			As fuel is used up, the decreased mass increases the acceleration due to equation: $a = \frac{F}{m}$		The acceleration increases as the forward thrust increases. The decrease in mass of fuel increases the forward thrust.		The rockets do not exert a downwards force on the Earth. The rocket moves upwards as a reaction to the downwards force of the exhaust gases.	
7	E	62	$d = v \times t$ $d = 3.0 \times 10^8 \times 6.0 \times 365.25 \times 24 \times 60 \times 60$ $d = 5.68 \times 10^{16} \text{ m}$					
8	D	62	Statement I - Incorrect		Statement II - Correct		Statement III - Correct	
			Neptune has the longest distance but not the largest mass		As the mass of the planet increases, the gravitation field strength increases for all planets		There is no relationship between distance from sun and gravitational field strength	
9	E	65	$Q = 0.12 \text{ C}$ $I = 8.0 \mu\text{A} = 8.0 \times 10^{-6} \text{ A}$ $t = ?$ $Q = I \times t$ $0.12 = 8.0 \times 10^{-6} \times t$ $t = \frac{0.12}{8.0 \times 10^{-6}} = 15000 \text{ s}$					

10	A	29	The resistance is the gradient of the line in a voltage-current graph. If the resistance of the lamp increases as the current increases, the gradient of the line increases and the line gets steeper as current increases.																		
11	B	61	$V_s = 18\text{V}$ $V_2 = ?$ $R_1 = 7.2\text{ k}\Omega = 7200\Omega$ $R_2 = 3.6\text{ k}\Omega = 3600\Omega$ $V_2 = \frac{R_2}{R_1 + R_2} \times V_s$ $V_2 = \frac{3600}{7200 + 3600} \times 18$ $V_2 = 6\text{ V}$																		
12	C	29	<table><tr><th>Total Resistance in Parallel branches</th><th>Total Resistance in Circuit</th><th>Current on Ammeter A_1</th></tr><tr><td>$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{20} + \frac{1}{20} = \frac{2}{20}$ $\therefore R_T = \frac{20}{2} = 10\Omega$</td><td>$R_T = R_1 + R_2 = 20 + 10 = 30\Omega$</td><td>$I = \frac{V}{R} = \frac{9.0}{30} = 0.3\text{ A}$</td></tr></table>	Total Resistance in Parallel branches	Total Resistance in Circuit	Current on Ammeter A_1	$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{20} + \frac{1}{20} = \frac{2}{20}$ $\therefore R_T = \frac{20}{2} = 10\Omega$	$R_T = R_1 + R_2 = 20 + 10 = 30\Omega$	$I = \frac{V}{R} = \frac{9.0}{30} = 0.3\text{ A}$												
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13	C	71	$P = 2.5\text{ W}$ $V = ?$ $R = 10\Omega$ $P = \frac{V^2}{R} \therefore 2.5 = \frac{V^2}{10} \therefore V^2 = 2.5 \times 10 = 25 \therefore V = \sqrt{25} = 5\text{ V}$																		
14	B	80																			
15	E	52	$E = 501\text{kJ} = 501000\text{J}$ $m = ?$ $l = 3.34 \times 10^5\text{ J kg}^{-1}$ $E = m \times l$ $501000 = m \times 3.34 \times 10^5$ $m = \frac{501000}{3.34 \times 10^5} = 1.5\text{ kg}$																		
16	C	65	☒ A pressure will increase as the volume is reduced due to Boyle's Law $p_1V_1 = p_2V_2$ ☒ B pressure will increase as the volume is reduced due to Boyle's Law $p_1V_1 = p_2V_2$ ☒ C pressure increase caused by more frequent collision of the gas particles with walls ☒ D pressure will increase as the volume is reduced due to Boyle's Law $p_1V_1 = p_2V_2$ ☒ E no change to the temperature means there is no increase in the kinetic energy																		
17	C	43	<table><tr><td>Volume: Constant</td><td>$p_1 = 1.3 \times 10^5\text{ Pa}$ $p_2 = ?$</td><td>$T_1 = 27^\circ\text{C} = 300\text{ K}$ $T_2 = 55^\circ\text{C} = 328\text{ K}$</td></tr><tr><td>$\frac{p_1}{p_2} =$</td><td>$\frac{T_1}{T_2}$</td><td></td></tr><tr><td>$\frac{1.3 \times 10^5}{p_2} =$</td><td>$\frac{300}{328}$</td><td></td></tr><tr><td>$p_2 \times 300 =$</td><td>$1.3 \times 10^5 \times 328$</td><td></td></tr><tr><td>$p_2 =$</td><td>$\frac{1.3 \times 10^5 \times 328}{300}$</td><td></td></tr><tr><td>$p_2 =$</td><td>$1.4 \times 10^5\text{ Pa}$</td><td></td></tr></table>	Volume: Constant	$p_1 = 1.3 \times 10^5\text{ Pa}$ $p_2 = ?$	$T_1 = 27^\circ\text{C} = 300\text{ K}$ $T_2 = 55^\circ\text{C} = 328\text{ K}$	$\frac{p_1}{p_2} =$	$\frac{T_1}{T_2}$		$\frac{1.3 \times 10^5}{p_2} =$	$\frac{300}{328}$		$p_2 \times 300 =$	$1.3 \times 10^5 \times 328$		$p_2 =$	$\frac{1.3 \times 10^5 \times 328}{300}$		$p_2 =$	$1.4 \times 10^5\text{ Pa}$	
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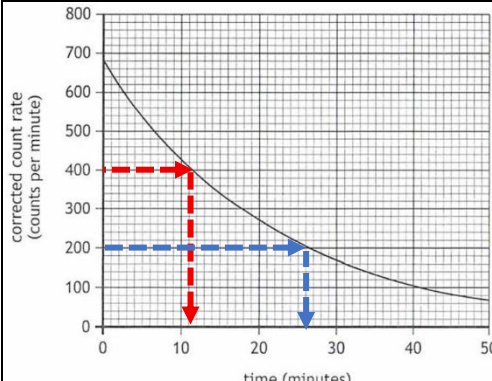
Question	Answer	Physics Covered														
1a(i)A	89m	Horizontal displacement = 52m + 28m = 80m Vertical displacement = 39m  $x = \sqrt{(80)^2 + (39)^2}$ $x = \sqrt{6400 + 1521}$ $x = \sqrt{7921}$ $x = 89\text{m}$														
1a(i)B	116	$\tan \theta = \frac{\text{opp}}{\text{adj}} = \frac{39}{80} = 0.4875 \therefore \theta = 26.0^\circ$ $\text{Bearing} = 90^\circ + 26^\circ = 116^\circ$														
1a(ii)	4.9 m s ⁻¹ at 116	$s = 89\text{ m} \qquad \qquad \qquad \bar{v} = ? \qquad \qquad \qquad t = 18\text{ s}$ $s = \bar{v} t \qquad (1\text{ mark})$ $89 = \bar{v} \times 18 \qquad (1\text{ mark})$ $\bar{v} = 89/18 = 4.9\text{ m s}^{-1} \qquad (1\text{ mark})$														
1b	192m	 <table><tr><th>Area ❶</th><th>Area ❷</th></tr><tr><td>Distance = area under graph</td><td>Distance = area under graph</td></tr><tr><td>$= \frac{1}{2} \times 10 \times 16$</td><td>$= 7 \times 16$</td></tr><tr><td>$= 80\text{m}$</td><td>$= 112\text{m}$</td></tr><tr><td colspan="2">Distance travelled = 80m + 112m = 192m</td></tr></table>	Area ❶	Area ❷	Distance = area under graph	Distance = area under graph	$= \frac{1}{2} \times 10 \times 16$	$= 7 \times 16$	$= 80\text{m}$	$= 112\text{m}$	Distance travelled = 80m + 112m = 192m					
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Distance travelled = 80m + 112m = 192m																
2a	Path shown as in diagram:															
2b(i)	0.70 s	$d = 9.1\text{ m} \qquad \qquad \qquad \bar{v} = 13\text{ m s}^{-1} \qquad \qquad \qquad t = ?$ $d = \bar{v} t \qquad (1\text{ mark})$ $9.1 = 13 \times t \qquad (1\text{ mark})$ $t = 9.1/13 = 0.70\text{ s} \qquad (1\text{ mark})$														
2b(ii)	6.9 m s ⁻¹	$a = 9.8\text{ m s}^{-2} \qquad \qquad \qquad v = ? \qquad \qquad \qquad u = 0\text{ m s}^{-1} \qquad \qquad \qquad t = 0.7\text{ s}$ $a = \frac{v - u}{t} \qquad (1\text{ mark})$ $9.8 = \frac{v - 0}{0.70} \qquad (1\text{ mark})$ $9.8 \times 0.7 = v - 0$ $v = 6.86\text{ m s}^{-1}$														
2c	Answer to include:	<table><tr><td>1st mark:</td><td>Same time</td></tr><tr><td>2nd mark:</td><td>Vertical/downward acceleration is the same</td></tr></table>	1 st mark:	Same time	2 nd mark:	Vertical/downward acceleration is the same										
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2 nd mark:	Vertical/downward acceleration is the same															
3	Open ended question: <table><tr><td>0 marks</td><td>1 marks</td><td>2 marks</td><td>3 marks</td></tr><tr><td>31%</td><td>39%</td><td>23%</td><td>7%</td></tr></table>	0 marks	1 marks	2 marks	3 marks	31%	39%	23%	7%	<table><tr><th>1 mark</th><th>2 marks</th><th>3 marks</th></tr><tr><td>Candidate has demonstrated a limited understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood at least a little of the physics within the problem.</td><td>Candidate has demonstrated a reasonable understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood the problem.</td><td>Candidate has demonstrated a good understanding of the physics involved. They show a good comprehension of the physics of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks.</td></tr></table>	1 mark	2 marks	3 marks	Candidate has demonstrated a limited understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood at least a little of the physics within the problem.	Candidate has demonstrated a reasonable understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood the problem.	Candidate has demonstrated a good understanding of the physics involved. They show a good comprehension of the physics of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks.
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4a	Explanation showing:	Direction Up ↑ One from: thrust rocket thrust engine thrust engine force Force from exhaust gases on rocket (1 mark)  (1 mark) Direction Down ↓ One from: weight gravitational pull pull of gravity force due to gravity		
4b(i)	Working as shown:	W = ? m = 5.5x10⁵ kg g = 9.8 N kg⁻¹ W = m x g (1 mark) W = 5.5x10⁵ x 9.8 (1 mark) W = 5.4x10⁶ N		
4b(ii)	7.7x10 ⁶ N	F = ? m = 5.5x10⁵ kg a = 4.1 m s⁻² F_{un} = m x a (1 mark) F_{un} = 5.5x10⁵ x 4.1 (1 mark) F_{un} = 2.3x10⁶ N F_{un} = Initial Upward Thrust - weight 2.3x10⁶ = Initial Upward Thrust - 5.4x10⁶ (1 mark) Initial Upward Thrust = 2.3x10⁶ + 5.4x10⁶ Initial Upward Thrust = 7.7x10⁶ N (1 mark)		
4c(i)	1.5 hours	 d = 43500 km s = 29000 km h⁻¹ t = ? d = s x t (1 mark) 43500 = 29000 x t (1 mark) t = 43500 / 29000 = 1.5 h (1 mark) Distance = 43500 km (1 mark)		
4c(ii)	16	no. of orbits in 1 day = $\frac{\text{hours in one day}}{\text{time for 1 orbit}} = \frac{24}{1.5} = 16$ orbits		
4d	Answer to include:	<table><tr><td>Friction (with the Earth's atmosphere)</td><td>kinetic energy is converted into (heat and) light (energy).</td></tr></table>	Friction (with the Earth's atmosphere)	kinetic energy is converted into (heat and) light (energy).
Friction (with the Earth's atmosphere)	kinetic energy is converted into (heat and) light (energy).			
5a	2.4 V	As switch S is open, electricity flows through two resistors in series. V_s = 6.0V V₂ = ? R₁ = 36Ω R₂ = 24Ω V₂ = $\frac{R_2}{R_1 + R_2} \times V_s$ (1 mark) V₂ = $\frac{24}{36 + 24} \times 6.0$ (1 mark) V₂ = 2.4 V (1 mark)		

5b(i)	42 Ω	As switch S in now closed, there are two 36Ω resistors in parallel attached in series to a 24Ω resistor.																												
		Combining Parallel Resistors:  $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} \quad (1 \text{ mark})$ $\frac{1}{R_T} = \frac{1}{36} + \frac{1}{36} \quad (1 \text{ mark})$ $\frac{1}{R_T} = \frac{2}{36}$ $R_T = \frac{36}{2} = 18 \Omega$	Combining Series Resistors:  $R_T = 18\Omega + 24\Omega \quad (1 \text{ mark})$ $R_T = 42 \Omega \quad (1 \text{ mark})$																											
5b(ii)	Answer to include:	<table><tr><td>1st mark:</td><td>Voltage across 24Ω resistor increases</td></tr><tr><td>2nd mark:</td><td>Closing switch S adds 36Ω resistor in parallel and combined resistance decreases</td></tr><tr><td>3rd mark:</td><td>Less voltage across parallel section increases voltage across 24Ω resistor.</td></tr></table>		1 st mark:	Voltage across 24Ω resistor increases	2 nd mark:	Closing switch S adds 36Ω resistor in parallel and combined resistance decreases	3 rd mark:	Less voltage across parallel section increases voltage across 24Ω resistor.																					
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6a	Direction of electron flow (continually) changing back and forth.	<table><tr><td>a.c.</td><td>Direction of electrons in current changes back and forth at regular intervals and size of the current varies with time and is not constant</td></tr><tr><td>d.c.</td><td>Negative charges (electrons) flow in one direction only and gives a constant trace on oscilloscope</td></tr></table>		a.c.	Direction of electrons in current changes back and forth at regular intervals and size of the current varies with time and is not constant	d.c.	Negative charges (electrons) flow in one direction only and gives a constant trace on oscilloscope																							
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6b(i)	Answer to include:	<table><tr><td>1st mark:</td><td>when the current (in the flex) is too high/great</td></tr><tr><td>2nd mark:</td><td>the fuse breaks/melts</td></tr></table>		1 st mark:	when the current (in the flex) is too high/great	2 nd mark:	the fuse breaks/melts																							
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6b(ii)	13A	<table><tr><td>P = 1500W</td><td>I = ?</td><td>V = 230V</td></tr><tr><td>P</td><td>=</td><td>I</td><td>V</td><td>(1 mark)</td></tr><tr><td>1500</td><td>=</td><td>I</td><td>x</td><td>230</td><td>(1 mark)</td></tr><tr><td>I</td><td>=</td><td colspan="2">1500/230 = 6.5 A</td><td>(1 mark)</td></tr><tr><td colspan="5">13A fuse is required (1 mark)</td></tr></table>		P = 1500W	I = ?	V = 230V	P	=	I	V	(1 mark)	1500	=	I	x	230	(1 mark)	I	=	1500/230 = 6.5 A		(1 mark)	13A fuse is required (1 mark)							
P = 1500W	I = ?	V = 230V																												
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1500	=	I	x	230	(1 mark)																									
I	=	1500/230 = 6.5 A		(1 mark)																										
13A fuse is required (1 mark)																														
6c(i)	Working showing:	<table><tr><td>E_h = ?</td><td>c = 4180 J kg⁻¹ °C⁻¹</td><td>m = 9.5 kg</td><td>ΔT= 65°C – 12°C = 53°C</td></tr><tr><td>E</td><td>=</td><td>c</td><td>x</td><td>m</td><td>x</td><td>ΔT</td><td>(1 mark)</td></tr><tr><td>E</td><td>=</td><td>4180</td><td>x</td><td>9.5</td><td>x</td><td>53</td><td>(1 mark)</td></tr><tr><td>E</td><td>=</td><td colspan="5">2.1x10⁶ J</td></tr></table>		E _h = ?	c = 4180 J kg ⁻¹ °C ⁻¹	m = 9.5 kg	ΔT= 65°C – 12°C = 53°C	E	=	c	x	m	x	ΔT	(1 mark)	E	=	4180	x	9.5	x	53	(1 mark)	E	=	2.1x10 ⁶ J				
E _h = ?	c = 4180 J kg ⁻¹ °C ⁻¹	m = 9.5 kg	ΔT= 65°C – 12°C = 53°C																											
E	=	c	x	m	x	ΔT	(1 mark)																							
E	=	4180	x	9.5	x	53	(1 mark)																							
E	=	2.1x10 ⁶ J																												
6c(ii)A	1400 s	<table><tr><td>P = 1.5kW = 1500W</td><td>E = 2.1x10⁶ J</td><td>t = ?</td></tr><tr><td>P</td><td>=</td><td>E</td><td>t</td><td>(1 mark)</td></tr><tr><td>1500</td><td>=</td><td>2.1x10⁶</td><td>t</td><td>(1 mark)</td></tr><tr><td>t</td><td>=</td><td>2.1x10⁶</td><td>1500</td><td>= 1400 s</td><td>(1 mark)</td></tr></table>		P = 1.5kW = 1500W	E = 2.1x10 ⁶ J	t = ?	P	=	E	t	(1 mark)	1500	=	2.1x10 ⁶	t	(1 mark)	t	=	2.1x10 ⁶	1500	= 1400 s	(1 mark)								
P = 1.5kW = 1500W	E = 2.1x10 ⁶ J	t = ?																												
P	=	E	t	(1 mark)																										
1500	=	2.1x10 ⁶	t	(1 mark)																										
t	=	2.1x10 ⁶	1500	= 1400 s	(1 mark)																									
6c(ii)B	One answer from:	<table><tr><td>Heat energy is lost (from the water) to the surroundings</td><td>Some of the energy is used to heat dishwasher/dishes/element</td></tr></table>		Heat energy is lost (from the water) to the surroundings	Some of the energy is used to heat dishwasher/dishes/element																									
Heat energy is lost (from the water) to the surroundings	Some of the energy is used to heat dishwasher/dishes/element																													

6d	Answer to include:	<table><tr><td>1st mark:</td><td colspan="2">Voltage across the thermistor decreases</td></tr><tr><td>2nd mark:</td><td>MOSFET transistor</td><td rowspan="2">} switches { off deactivates</td></tr><tr><td>3rd mark:</td><td>Relay switches</td></tr></table>	1 st mark:	Voltage across the thermistor decreases		2 nd mark:	MOSFET transistor	} switches { off deactivates	3 rd mark:	Relay switches							
1 st mark:	Voltage across the thermistor decreases																
2 nd mark:	MOSFET transistor	} switches { off deactivates															
3 rd mark:	Relay switches																
7a	Answer to include:	<table><tr><td>1st mark:</td><td colspan="2">Graph does not go through origin so not directly proportional</td></tr><tr><td>2nd mark:</td><td colspan="2">Kelvin scale should be used to get graph to go through origin</td></tr></table>	1 st mark:	Graph does not go through origin so not directly proportional		2 nd mark:	Kelvin scale should be used to get graph to go through origin										
1 st mark:	Graph does not go through origin so not directly proportional																
2 nd mark:	Kelvin scale should be used to get graph to go through origin																
7b	Gas particles move more slowly	As temperature of the gas decreases, the gas particles have less (kinetic) energy and gas particles collide less with the walls of the container. The volume of gas decreases as the gas particles move closer together															
7c	One answer from:	<table><tr><td>So that the gas is heated evenly</td><td>So that the temperature of the water is evenly distributed.</td></tr></table>	So that the gas is heated evenly	So that the temperature of the water is evenly distributed.													
So that the gas is heated evenly	So that the temperature of the water is evenly distributed.																
7d	0.084 N	<table><tr><td>$P = 1.02 \times 10^5 \text{ Pa}$</td><td>$F = ?$</td><td>$A = 0.82 \times 10^{-6} \text{ m}^2$</td></tr><tr><td></td><td>$P = \frac{F}{A}$</td><td>(1 mark)</td></tr><tr><td></td><td>$1.02 \times 10^5 = \frac{F}{0.82 \times 10^{-6}}$</td><td>(1 mark)</td></tr><tr><td></td><td>$F = 1.02 \times 10^5 \times 0.82 \times 10^{-6}$</td><td></td></tr><tr><td></td><td>$F = 0.084 \text{ N}$</td><td>(1 mark)</td></tr></table>	$P = 1.02 \times 10^5 \text{ Pa}$	$F = ?$	$A = 0.82 \times 10^{-6} \text{ m}^2$		$P = \frac{F}{A}$	(1 mark)		$1.02 \times 10^5 = \frac{F}{0.82 \times 10^{-6}}$	(1 mark)		$F = 1.02 \times 10^5 \times 0.82 \times 10^{-6}$			$F = 0.084 \text{ N}$	(1 mark)
$P = 1.02 \times 10^5 \text{ Pa}$	$F = ?$	$A = 0.82 \times 10^{-6} \text{ m}^2$															
	$P = \frac{F}{A}$	(1 mark)															
	$1.02 \times 10^5 = \frac{F}{0.82 \times 10^{-6}}$	(1 mark)															
	$F = 1.02 \times 10^5 \times 0.82 \times 10^{-6}$																
	$F = 0.084 \text{ N}$	(1 mark)															
8a	Diagram showing:																
8b(i)	Line graph showing:	<table><tr><td><u>1 mark</u> suitable scales, labels and units</td><td><u>1 mark</u> all points plotted accurately to $\pm$ half a division</td><td><u>1 mark</u> best fit straight line</td></tr><tr><td> Best fit straight line</td><td colspan="2"> Best fit curve</td></tr></table>	<u>1 mark</u> suitable scales, labels and units	<u>1 mark</u> all points plotted accurately to $\pm$ half a division	<u>1 mark</u> best fit straight line	 Best fit straight line	 Best fit curve										
<u>1 mark</u> suitable scales, labels and units	<u>1 mark</u> all points plotted accurately to $\pm$ half a division	<u>1 mark</u> best fit straight line															
 Best fit straight line	 Best fit curve																

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10c(ii)	65661 Hz Accept 65660 Hz or 65700 Hz 66000 Hz or 70000 Hz	$f_w = \left[\frac{v}{v - v_{bat}} \right] \times f_{bat}$ $f_w = \left[\frac{340}{340 - 8.6} \right] \times 64000 \quad (1 \text{ mark})$ $f_w = \left[\frac{340}{331.4} \right] \times 64000$ $f_w = 65661 \text{ Hz} \quad (1 \text{ mark})$																		
11a	<u>1 mark</u> Beta particles will be absorbed by aluminium <u>1 mark</u> Alpha radiation will be absorbed by air.	<table><tr><td>Type of Radiation</td><td>Alpha</td><td>Beta</td><td>Gamma</td></tr><tr><td>Stopped by</td><td>piece of paper or few cm of air</td><td>layer of Aluminium</td><td>layer of lead</td></tr></table>	Type of Radiation	Alpha	Beta	Gamma	Stopped by	piece of paper or few cm of air	layer of Aluminium	layer of lead										
Type of Radiation	Alpha	Beta	Gamma																	
Stopped by	piece of paper or few cm of air	layer of Aluminium	layer of lead																	
11b(i)	One answer from:	<table><tr><td>The count rate/reading will include the source and background (count)</td><td>Background (count rate) will need to be subtracted</td><td>To determine the count rate/reading due to the source (alone).</td></tr></table>	The count rate/reading will include the source and background (count)	Background (count rate) will need to be subtracted	To determine the count rate/reading due to the source (alone).															
The count rate/reading will include the source and background (count)	Background (count rate) will need to be subtracted	To determine the count rate/reading due to the source (alone).																		
11b(ii)A	15 minutes	 Take any halving of the corrected count rate on y-axis. It should give the same time interval on the x-axis. e.g. Halving the corrected count rate from 400 to 200 time at 200 cpm: 26 minutes time at 400 cpm: 11 minutes Half-life: 26–11 = 15 minutes																		
11b(ii)B	<u>1 mark:</u> 4 half-lives <u>1 mark:</u> 60 minutes	<table><tr><td>Count Rate Fraction</td><td>1</td><td>$\frac{1}{2}$</td><td>$\frac{1}{4}$</td><td>$\frac{1}{8}$</td><td>$\frac{1}{16}$</td></tr><tr><td>No. of half-lives</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>Time (minutes)</td><td>0</td><td>15</td><td>30</td><td>45</td><td>60</td></tr></table>	Count Rate Fraction	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	No. of half-lives	0	1	2	3	4	Time (minutes)	0	15	30	45	60
Count Rate Fraction	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$															
No. of half-lives	0	1	2	3	4															
Time (minutes)	0	15	30	45	60															
11b(iii)	One answer from:	<table><tr><td>Limit time of exposure</td><td>Increase distance from source</td><td>Place shielding in the way</td></tr></table>	Limit time of exposure	Increase distance from source	Place shielding in the way															
Limit time of exposure	Increase distance from source	Place shielding in the way																		
12a	One answer from:	<table><tr><td>Activity (of the source) decreases (with time)</td><td>Fewer radioactive particles emitted per second</td></tr></table>	Activity (of the source) decreases (with time)	Fewer radioactive particles emitted per second																
Activity (of the source) decreases (with time)	Fewer radioactive particles emitted per second																			
12b(i)	Fast moving or High energy electron	<table><tr><td>Type of Radiation</td><td>Alpha</td><td>Beta</td><td>Gamma</td></tr><tr><td>Nature</td><td>Helium Nucleus</td><td>Fast Moving Electron from nucleus</td><td>Electromagnetic radiation</td></tr></table>	Type of Radiation	Alpha	Beta	Gamma	Nature	Helium Nucleus	Fast Moving Electron from nucleus	Electromagnetic radiation										
Type of Radiation	Alpha	Beta	Gamma																	
Nature	Helium Nucleus	Fast Moving Electron from nucleus	Electromagnetic radiation																	
12b(ii)A	$1.54 \times 10^{-13} \text{ J}$	$1 \text{ MeV} \longleftrightarrow 1.60 \times 10^{-13} \text{ J}$ $0.960 \text{ MeV} \longleftrightarrow 1.60 \times 10^{-13} \text{ J} \times \frac{0.960}{1}$ $= 1.54 \times 10^{-13} \text{ J}$																		
12b(ii)B	1.2 Gy	$D = ? \quad E = 0.0768 \text{ J} \quad m = 0.064 \text{ kg}$ $D = \frac{E}{m} = \frac{0.0768}{0.064} = 1.2 \text{ Gy}$ (1 mark) (1 mark) (1 mark)																		
12b(iii)	1.2 Sv	$H = ? \quad D = 1.2 \text{ Gy} \quad W_r = 1$ <table><tr><td>H</td><td>=</td><td>D</td><td>x</td><td>W_r</td><td>(1 mark)</td></tr><tr><td>H</td><td>=</td><td>1.2</td><td>x</td><td>1</td><td>(1 mark)</td></tr><tr><td>H</td><td>=</td><td>1.2 Sv</td><td></td><td></td><td>(1 mark)</td></tr></table>	H	=	D	x	W_r	(1 mark)	H	=	1.2	x	1	(1 mark)	H	=	1.2 Sv			(1 mark)
H	=	D	x	W_r	(1 mark)															
H	=	1.2	x	1	(1 mark)															
H	=	1.2 Sv			(1 mark)															

2026 National 5 Physics Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	E	Missing	Invalid
1	A	81	10	1	6	2	1	0
2	D	6	4	7	69	13	2	0
3	D	1	1	14	24	60	1	0
4	C	4	5	69	10	11	1	0
5	B	1	93	1	4	0	1	0
6	D	12	7	15	24	41	1	0
7	E	11	14	5	7	62	1	0
8	D	2	26	7	62	2	1	0
9	E	10	8	13	3	65	1	0
10	A	29	4	53	8	5	1	0
11	B	6	61	14	9	10	1	0
12	C	38	8	29	20	3	2	0
13	C	11	5	71	12	0	1	0
14	B	11	80	1	4	2	1	0
15	E	8	15	13	9	52	2	0
16	C	2	4	65	25	3	1	0
17	C	4	4	43	40	7	1	0
18	A	15	52	3	23	6	1	0
19	C	3	8	76	3	10	1	0
20	A	74	2	13	6	4	1	0
21	A	72	2	5	15	5	1	0
22	E	7	10	5	11	66	1	0
23	E	6	32	11	24	24	2	0
24	B	15	53	6	20	4	2	0
25	D	5	5	16	53	20	1	0

2026 National 5 Physics Section 2 Marks Data Analysis

2026 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1aiA	2	1.3	66.9%	54	3.7%
Q1aiB	2	0.9	46.1%	62	7.5%
Q1aii	3	1.5	50.9%	52	3.7%
Q1b	3	1.7	56.4%	55	2.0%
Q2a	1	0.6	61.1%	32	1.7%
Q2bi	3	2.7	88.4%	30	1.8%
Q2bii	3	1.2	39.0%	63	14.3%
Q2c	2	0.5	24.1%	45	3.9%
Q3	3	1.1	35.7%	62	4.8%
Q4a	2	1.2	59.7%	54	1.6%
Q4bi	2	1.7	82.9%	46	3.3%
Q4bii	4	1.9	48.5%	65	7.4%
Q4ci	3	1.7	57.3%	55	3.9%
Q4cii	1	0.6	59.5%	52	8.0%
Q4d	2	0.5	26.0%	36	5.6%
Q5a	3	1.7	57.9%	62	5.5%
Q5bi	4	2.5	62.8%	63	6.3%
Q5bii	3	0.9	28.4%	53	10.2%
Q6a	1	0.3	27.4%	43	4.6%
Q6bi	2	0.8	39.6%	48	6.4%
Q6bii	4	2.5	62.6%	56	9.2%
Q6ci	2	1.1	56.5%	50	6.2%
Q6ciiA	3	1.6	52.6%	68	20.3%
Q6ciiB	1	0.5	53.2%	56	14.7%
Q6d	3	0.7	25.0%	58	13.8%
Q7a	2	0.4	18.5%	52	10.0%
Q7b	1	0.5	50.8%	36	4.0%
Q7c	1	0.3	30.5%	30	10.3%
Q7d	3	2.3	75.5%	58	9.9%
Q8a	1	0.3	30.8%	34	4.4%
Q8bi	3	1.8	59.6%	40	1.8%
Q8bii	1	0.7	72.5%	35	3.7%
Q8c	4	1.9	47.6%	69	10.3%
Q8di	3	1.7	57.0%	70	8.5%
Q8dii	2	0.5	25.3%	2	13.4%
Q9	3	0.8	25.7%	64	16.1%
Q10a	4	2.7	67.2%	66	7.0%
Q10b	1	0.4	38.5%	42	12.8%
Q10ci	1	0.3	33.2%	54	9.0%
Q10cii	2	1.1	56.6%	61	9.7%
Q11a	2	0.5	23.3%	46	7.7%
Q11bi	1	0.5	49.1%	40	9.6%
Q11biiA	1	0.6	62.9%	49	5.7%
Q11biiB	2	1.1	54.2%	59	12.1%
Q11biii	1	0.6	62.3%	28	5.5%
Q12a	1	0.4	40.1%	40	5.9%
Q12bi	1	0.4	44.3%	48	9.6%
Q12biiA	1	0.5	51.5%	47	11.5%
Q12biiB	3	1.2	40.6%	64	11.8%
Q12biii	3	1.9	63.1%	70	15.5%
Paper	110	55.2	50.2%		

2026 National 5 Physics Section 2 Frequency of Marks Analysis

2026 Item	0	1	2	3	4
Q1aiA	30	6	64		
Q1aiB	42	24	34		
Q1aii	34	3	38	24	
Q1b	22	32	2	44	
Q2a	39	61			
Q2bi	7	4	4	84	
Q2bii	49	16	2	32	
Q2c	64	25	12		
Q3	31	39	23	7	
Q4a	25	31	44		
Q4bi	8	18	74		
Q4bii	16	7	60	2	15
Q4ci	12	41	11	37	
Q4cii	40	60			
Q4d	53	43	5		
Q5a	20	32	2	46	
Q5bi	32	2	3	8	55
Q5bii	56	17	12	14	
Q6a	73	27			
Q6bi	46	28	25		
Q6bii	30	8	2	4	57
Q6ci	15	58	27		
Q6ciiA	42	6	5	47	
Q6ciiB	47	53			
Q6d	55	23	13	8	
Q7a	68	26	5		
Q7b	49	51			
Q7c	69	31			
Q7d	15	4	20	60	
Q8a	69	31			
Q8bi	8	18	62	12	
Q8bii	27	73			
Q8c	17	45	2	3	33
Q8di	18	19	36	27	
Q8dii	67	16	17		
Q9	48	31	15	5	
Q10a	11	12	24	2	51
Q10b	61	39			
Q10ci	67	33			
Q10cii	30	27	43		
Q11a	60	34	6		
Q11bi	51	49			
Q11biiA	37	63			
Q11biiB	41	10	49		
Q11biii	38	62			
Q12a	60	40			
Q12bi	56	44			
Q12biiA	49	51			
Q12biiB	22	54	5	19	
Q12biii	26	11	11	52	

2026 National 5 Physics Section 3 Assignment Marks Analysis

Item	Max Mark	Average Mark	% Average Mark	Discrimination
1	1	0.9	88.5%	24
2	3	1.5	50.2%	43
3a	1	0.6	58.4%	29
3b	1	0.9	92.9%	16
3c	1	0.7	66.0%	32
3d	1	0.7	65.9%	32
3e	1	0.9	92.9%	29
3f	1	0.9	89.0%	26
4a	1	1	97.0%	20
4b	1	0.9	89.3%	26
4c	1	0.9	89.8%	26
4d	1	0.3	29.6%	29
5	1	0.6	55.6%	38
6	1	0.6	58.1%	32
7	2	0.5	27.2%	21
8a	1	1	97.0%	14
8b	1	1	98.5%	17
Paper	20	13.7	68.7%	

Item	Max Mark	0	1	2	3
1	1	11	89		
2	3	11	42	33	14
3a	1	42	58		
3b	1	7	93		
3c	1	34	66		
3d	1	34	66		
3e	1	7	93		
3f	1	11	89		
4a	1	3	97		
4b	1	11	89		
4c	1	10	90		
4d	1	70	30		
5	1	44	56		
6	1	42	58		
7	2	68	10	22	
8a	1	3	97		
8b	1	1	99		
Paper	20				

Item	Max Mark	Item Description
1	1	An aim that describes clearly the purpose of the investigation.
2	3	An account of physics relevant to the aim of the investigation.
3a	1	A brief description of the approach used to collect experimental data.
3b	1	Sufficient raw data from the candidate's experiment.
3c	1	Data presented in a correctly produced table
3d	1	Mean and/or derived values calculated correctly
3e	1	Data relevant to the experiment from an internet/literature source
3f	1	A reference for the source of the internet/literature data
4a	1	An appropriate format from the options of scatter, line or bar graph.
4b	1	The axis/axes of the graph has/have suitable scales.
4c	1	The axes of the graph have suitable labels and units
4d	1	Accurately plotted data points and, where appropriate, a line of best fit
5	1	Valid comparison of the experimental data with data from internet/literature source.
6	1	A conclusion that relates to aim and supported by all the data in the report.
7	2	An evaluation of the experimental procedure.
8a	1	An informative title.
8b	1	A clear and concise report.